

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



December 13, 2006

MATERIALS LABORATORY FACTUAL REPORT

Report No. 06-073

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : Connector tunnel to Ted Williams Tunnel
NTSB No. : HWY06MA024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Core samples from the westbound tunnel (three samples with attached anchors, two samples with the anchors removed).

C. DETAILS OF THE EXAMINATION

1. GENERAL DESCRIPTION

A total of 10 anchor positions in the ceiling of the westbound connector tunnel were selected for removal of core samples. The samples were removed on August 9, 2006 using cylindrical diamond coring bits.¹ Water flowing through the center of the bit provided cooling and flushing at the cutting interface. The anchors were extracted from six of the positions prior to cutting of the cores, and the maximum reaction load required to extract the anchors was measured using a hand-operated hydraulic press. The table on the following page lists the anchor positions from which core samples were removed, the outside diameter of the removed core section, and the maximum reacted load for those anchors that were removed. The final column indicates "NTSB" if the NTSB Materials Laboratory retained the core sample or "OAG" if the Massachusetts Office of the Attorney General retained the core sample.

The remainder of this section contains a general description of the core samples retained by the NTSB. Unless otherwise mentioned, all rebar found in the core samples was aligned approximately horizontally. Rebar will be referenced using its designation number, with #5 rebar being nominally 5/8 inch in diameter, #6 rebar being nominally 3/4 inch in diameter, and #11 rebar being nominally 1 3/8 inches in diameter.

¹ The upper half of core sample "WB083-N-1 north" was removed on September 27, 2006.

Core Sample Information				
Anchor Position		Diameter	Maximum Load	Retained By
WB084-N-2	north	3.2 inches	13,810 pounds	OAG
	south	3.2 inches	16,213 pounds	OAG
WB084-S-3	north	3.2 inches	15,993 pounds	OAG
	south	3.2 inches	15,769 pounds	NTSB
WB083-S-8	north	3.7 inches	N/A	NTSB
	south	3.7 inches	N/A	OAG
WB083-N-3	north	3.7 inches	N/A	NTSB
	south	3.7 inches	N/A	NTSB
WB083-N-1	north	3.2 inches	13,000 pounds	NTSB
	south	3.2 inches	13,996 pounds	OAG

Examination, inspection, and sectioning of the core samples described in this report revealed the presence of significant voids (unbonded surfaces) in the epoxy between the anchors and the concrete surface of the anchor hole. In general, the surfaces of the void areas and the epoxy surfaces that separated cleanly from the threads of an anchor were very smooth with a greenish tint. In locations where void areas were deformed because they were adjacent to fractured or displaced epoxy, the color of the void surface changed to white, approximately the color of the fractured regions of epoxy. All of the voids documented appeared to be wholly contained within the epoxy layer, i.e., the voids did not extend directly to either the concrete surface of the anchor hole or to the anchor. Descriptions of the as-received core samples are contained in the remainder of this section. A discussion of the results of X-ray inspection of the cores is contained in Section 2 of this report. Descriptions and photographs of the voids are located in Section 3, and measurements of the extent of the voids are summarized in Section 4.

1.1 WB084-S-3 south

Figure 1 shows a side view and an oblique view of the core sample at the “WB084-S-3 south” anchor position. The anchor was removed prior to cutting of the core sample, and had not been displaced downward prior to removal. This sample was a total of 7 inches high, and the concrete was fractured at about 3.5 inches from the ceiling intersecting a piece of #6 rebar. The anchor hole was cored almost directly into the center of a #11 rebar. The lower edge of the #11 rebar was about 4 inches above the ceiling, and, based on a typical anchor hole depth of more than 5 inches, the top of the anchor hole would have been at or just above the top of this piece of rebar. The fracture at the top of the core sample intersected a #6 rebar whose lower edge was 6.6 inches above the ceiling. Also contained in the core sample was the end of a piece of #5 rebar at the location indicated in figure 1.

1.2 WB083-S-8 north

Figure 2 shows a side view and an oblique view of the core sample at the “WB083-S-8 north” anchor position. The anchor had not been displaced downward prior to removal of the core sample. This sample was about 6.5 inches high, and the anchor had not been displaced downward. The edge of the anchor hole was about 0.6 inch from a #11 rebar. The bottom of this rebar was about 1.3 inch above the ceiling. The anchor hole did appear to largely intersect a #5 rebar that was just above and perpendicular to the #11 rebar. The bottom of this #5 rebar was 2.75 inch above the ceiling. A second #5 rebar, oriented parallel to the first, was noted in the core sample with its bottom edge 5.75 inch above the ceiling. The top of the core sample intersected a second layer of #11 rebar, with its lower edge 6.4 inches above the ceiling. This #11 rebar was parallel to but about 1 inch laterally offset from the lower #11 rebar.

1.3 WB083-N-3 north

Figure 3 shows a side view and an oblique view of the core sample at the “WB083-N-3 north” anchor position. The anchor had not been displaced downward prior to removal of the core sample. This sample was about 4.5 inches high, and had about ½ inch of the anchor protruding from the upper end of the core sample. The upper end of the core sample was fractured around a #11 rebar, and the anchor hole was drilled almost directly through this bar. The bottom of the #11 rebar was 3.2 inches above the ceiling.

1.4 WB083-N-3 south

Figure 4 shows a side view and an oblique view of the core sample at the “WB083-N-3 south” anchor position. This sample was about 4.5 inches high, and measurements of the distance between the cap flange and the ceiling surface on this sample indicated that the anchor had displaced downward 0.3 inch to 0.4 inch². The upper end of the core sample was fractured around a #11 rebar, and the hole for the anchor partially intersected this bar. The bottom of the #11 rebar was 3.1 inches above the ceiling.

1.5 WB083-N-1 north

Figure 5 shows a side view and an oblique view of the lower portion of the core sample at the “WB083-N-1 north” anchor position. The anchor was removed prior to cutting of the core sample, and had not been displaced downward prior to removal. During the removal from the ceiling, this sample fractured about 3 inches above the ceiling. The fracture was largely above a piece of #11 rebar and an end piece (slightly skewed from horizontal) of #5 rebar. The lower edge of the #11 rebar was 1.7 inches above the ceiling. The drilled hole for the anchor intersected a small amount of the #11 rebar and was mostly through the #5 rebar. The lower portion of the core sample also contained another piece of #5 rebar that was approximately parallel to the piece at the fracture. The upper portion of

² On-scene measurements, taken on a mechanical lift as part of the early on-scene documentation, indicated that the anchor had been displaced downward about 1/4 inch.

the core sample extended above the top of the anchor hole and did not contain or intersect any additional rebar pieces.

2. X-RAY INSPECTION

Following the initial overall examination described above, the core samples containing anchors (“WB083-S-8 north”, “WB083-N-3 north”, and “WB083-N-3 south”) were transported to the laboratory facilities of the Federal Bureau of Investigation, Quantico, Virginia, on August 22, 2006, where x-ray inspection of the core samples was conducted. The purpose of the inspection was to determine if any voids in the epoxy were visible in the three samples still containing anchors. X-ray inspection was conducted using a Hytec, Inc. FCT-3200 320 KV machine. Each sample was inspected from at least six different orientations (usually an arbitrarily defined 0°, then 30°, 60°, 90°, 120°, and 150°).

The x-rays images were able to show the sides of the drilled holes, but did not show clear evidence of voids in the epoxy between the threads of the anchors and the hole surface, although in some views, slightly darker bands suggestive of voids were visible. In several of the views of “WB083-S-8 north” what appeared to be an irregular void area was visible above the upper end of the anchor. Figure 6 shows the x-ray view with the most distinctly appearing void area.

3. SECTIONING OF THE CORE SAMPLES

3.1 WB084-S-3 south

The loosely adhering epoxy was removed from the lower portion of core sample “WB 084-S-3 south” using a slender wooden rod. It was noted that about one half of the circumference of the hole contained very little epoxy, and that significant epoxy remained on the other half of the hole. After removal of the loose epoxy, this portion of the sample was cut longitudinally in the Safety Board’s Materials Laboratory using a high-speed, water-cooled abrasive cut-off wheel. The cut was oriented such that one half would have the bulk of the remaining epoxy. Figure 7 shows the lower portion of this section after cutting, and figure 8 shows a closer view of the side with the bulk of the remaining epoxy. Multiple void areas were visible, and are outlined by the dashed lines in figure 8. In areas where most of the epoxy was missing from the surface of the drilled hole, the aggregate (a darker stone) was easily discernable from the mortar.

The saw cut surface visible in figure 7 was nearly through the centerline of the hole. It was noted that the intersections between the saw cut surface and the edges of the hole were nearly straight lines, and that the hole surface (where visible) contained no discernable cut lines or significant irregularities that might be expected when using a carbide drill bit. The straightness of the hole and its smooth surface appeared typical of a hole drilled with a diamond coring bit³.

³ Although longitudinal saw cuts were not made directly into the anchor hole in the four other core samples, the surface of these other holes, where visible, was also smooth and typical of a hole drilled with a diamond coring bit.

The portion of core sample “WB084-S-3 south” shown on the right in figure 7 was sent to Laser Techniques Company, Bellevue, Washington, so that an image of the interior of the hole could be generated with laser scanning equipment.⁴ Figure 9 shows the laser-generated image.

Examination of the “WB084-S-3 south” anchor showed that one side of the anchor contained a damaged and deformed layer of epoxy that appeared to have extended out to the surface of the anchor hole. The epoxy along this side of the anchor, shown in figure 10, appeared to have been compressed along the length of the anchor, with the relatively smooth surface of the epoxy apparently corresponding to the surface of the anchor hole. As can be seen in figure 10, the smoother surface of the epoxy in this area contained a reddish tint, not found on any other anchor that was a part of this examination. Portions of the epoxy with the reddish tint and without the reddish tint were subjected to X-ray energy dispersive spectroscopy in a scanning electron microscope. Spectra from the two areas were very similar, both showing peaks of calcium, silicon, and magnesium (associated with the filler material for the epoxy) as well as oxygen and carbon. Neither spectrum showed any amount of iron.

As can be seen in figure 1, core sample “WB084-S-3 south” extended above the top of the drilled anchor hole. In order to expose the top of the drilled hole, lateral cuts were made through the upper portion of the core sample just into or just short of the upper edge of the #11 rebar. Figure 11 shows the upper portion of this section after the concrete was fractured away from the upper surface of this rebar. The upper end of the anchor hole was contained in the piece shown on the left in this figure. As can be seen in figure 11, a large void was found in the epoxy located above the upper end of the anchor.

In order to more easily show the characteristics of the upper end of the anchor hole, the piece of core sample “WB084-S-3 south” shown on the left in figure 11 was heated to 100° C and held for 1 hour to remove moisture from the sample, then heated to approximately 300° C and held for 1 hour. This thermal exposure degraded the epoxy in the upper end of the hole, and the epoxy was then scraped out using a slender wooden rod. A closer view of the upper end of the hole is shown in figure 12. As can be seen in this figure, the perimeter of the hole was deeper than the central portion, which appeared to be the remnant of a fractured $\frac{3}{4}$ inch diameter core section created when the anchor hole was cut. Measurement of the depth of the drilled hole showed that it extended about 5.54 inches above the ceiling of the tunnel, leaving a space of about 0.35 inch above the upper end of the anchor. The physical characteristics of the upper end of the hole were consistent with creation of the hole using a diamond coring bit.

Three additional core samples, “WB083-S-8 north”, “WB083-N-3 north”, and “WB083-N-3 south”, were opened using cutting equipment at the Federal Highway Administration’s Turner Fairbank Highway Research Center.

⁴ The image was generated using a rotatable, translatable laser, designed and built by Laser Techniques Company, that was capable of generating an image of the entire interior surface of a similar hole without cutting the hole open. It was expected that the quality and characteristics of images generated from intact anchor holes would be similar to the quality and characteristics of image generated from the opened hole.

3.2 WB083-S-8 north

As shown in figure 2, core sample “WB083-S-8 north” contained the full length of the anchor hole, as well as some concrete above the top of the hole. This sample was cut longitudinally on diametrically opposite sides using a water-cooled saw from the outside diameter of the core sample up to but not into the anchor hole. The cuts were made such that the interior pieces of rebar would not hold the two halves together.⁵ After the initial longitudinal cuts were made, chisels were hammered into one of the cuts in attempts to fracture the remaining concrete between the saw cuts and the anchor hole. This process produced fractures that in some areas did not cleanly intersect the anchor hole. For this reason, an additional cut, at 90° to the previous cuts, was made through the larger of the two broken out pieces. The anchor, which had not displaced relative to the anchor hole was then pried off one of the smaller pieces of the core sample.

Initial examination of the pieces of the opened section showed that the epoxy filled most of the volume between the anchor and the hole, with the exception of one large void (approximately cylindrical area of 0.7 square inch) and some small voids. Figure 13 shows one side of the anchor and the mating side of the anchor hole with the large void outlined by arrows and individual smaller voids indicated by individual arrows. In addition, a large void and several smaller voids were noted at and above the top of the anchor, as shown in figure 14, consistent with what appeared to be a void at this location visible on the x-ray, see figure 6. At various locations on the pieces of the section, fractures occurred in the concrete adjacent to the epoxy, through the epoxy in areas without voids, and at the interface between the anchor and the epoxy. In one location, a piece of epoxy was noted that remained adhered to the cored surface of a piece of rebar that intersected the anchor hole. Also, pieces of concrete remained attached to the anchor, as can be seen in figure 13.

Measurement of the depth of the drilled hole showed that it extended 5.42 inches above the ceiling of the tunnel, leaving a space of about 0.4 inch above the upper end of the anchor. The upper end of the hole had characteristics also consistent with creation of the hole using a diamond coring bit.

3.3 WB083-N-3 north

As shown in figure 3, the hole for anchor “WB083-N-3 north” had been drilled nearly directly through a #11 rebar, and the core sample from this location fractured just above this rebar piece. Based on the difficulty encountered in cutting through the rebar on the previous sample, it was decided to make a transverse cut completely through this section just at the bottom of the #11 rebar piece, and then continue with longitudinal sectioning of the lower portion of the sample, using a process similar to that used for sample “WB083-S-8 north”. The pieces of core sample “WB083-N-3 north” after this cutting are shown in figure 15. The anchor, which had not displaced relative to the anchor hole, was removed

⁵ One of the initial longitudinal cuts on core sample WB083-S-8 north was perpendicular to the #11 rebar piece visible in figure 5, cutting this rebar piece in half. Both of the initial longitudinal cuts were parallel to the #5 rebar pieces, such that these rebar pieces would be on one half of the split sample after opening.

from the sample half to which it remained attached in order to also examine that portion of the epoxy. The maximum extent of the voids was revealed by chipping away the epoxy on the anchor side of void areas. Examination of the opened section revealed a single continuous void, as shown in figure 16.

It was also noted that void areas extended up into the area where the anchor passed through the rebar. Figure 17 shows a closer view of the anchor at the transverse cut located at the lower edge of the rebar. The void areas extending further up are indicated in this figure. Also, it was noted that the anchor was not concentrically located within the hole, and that the voids were more predominant on the side of the anchor with greater distance between the anchor and the hole surface. Voids were also noted on the same side of the anchor where the anchor extended out of the upper edge of the rebar piece.

3.4 WB083-N-3 south

As shown in figure 4, anchor "WB083-N-3 south" had displaced downward 0.3 inch to 0.4 inch, and the hole for this anchor had been drilled partially through a #11 rebar. Because the rebar did not totally surround the anchor hole, a transverse cut was made through the concrete up to the lower edge of the rebar piece, and the concrete between the cut and rebar was broken out. The rebar piece was then pulled laterally off the anchor. The remainder of the core sample was then cut longitudinally. Figure 18 shows the resulting pieces.

The anchor was removed from the sample half to which it remained attached in order to also examine that portion of the epoxy. The two left images in figure 19 (photographs "A" and "B") show closer views of the section halves after removal of the anchor. The two right images in figure 19 (photographs "C" and "D") show views of the section halves after chipping away epoxy to show the full extent of the voids.⁶

During the removal of epoxy to show the full extent of the voids, a large piece of epoxy with the imprint of the anchor threads separated from one half of the sample (compare photographs "A" and "C" in figure 19). On the other half of the sample, the epoxy was chipped away where it protruded over void areas, but no bigger pieces of epoxy were easily separated from the sample half, compare photographs "B" and "D" in figure 19.

Examination of the separated surfaces in photograph "C" in figure 19 revealed by the removal of the larger pieces of epoxy from the core sample half showed that the separation was through the epoxy, leaving layers of epoxy on the hole and anchor surfaces. In some areas, the separation surface was very smooth, and in other areas, the separation surface contained circumferential textured bands in a pattern roughly corresponding to the thread spacing on the anchor. Figure 20 shows a closer view of separated epoxy in image "C" of figure 19. Arrow "S" in figure 20 indicates a smooth portion of the epoxy surface, and the textured bands are visible in the left side of this photograph. Figure 21 shows a view of the

⁶ Because the anchor at position "WB083-N-3 south" had moved downward from its installed position, the full extent of void areas could have been masked by epoxy that moved with the anchor but remained in the hole after the anchor was removed from the split halves.

fractured edge of a piece of the epoxy removed from the area shown in image “A” of figure 19. As can be seen in this photograph, multiple cracks extended from the crowns of the threads into the epoxy in a portion of the piece that also had the textured bands. Detailed examination of the edge of the epoxy pieces in areas without the textured bands revealed no evidence of cracking between the thread crowns and the cylindrical separation surface.

Figure 22 shows additional closer views of the epoxy remaining in the anchor hole of the core sample. The lower left view shows the area in the box from image “D” of figure 19. In this area, multiple cracks that were approximately parallel with the hole surface were noted in the epoxy between the anchor and hole surface. In the area shown, these cracks did not intersect the threads of the anchor. The lower right view of figure 22 shows the area in the box from image “C” of figure 19. In this area multiple cracks were also noted. The upper right view of figure 22 shows the same area as the lower right view, but the direction of viewing is downward instead of tangential to the hole surface. The multiple crack planes are also visible in this image.

Detailed examination of the epoxy on core sample “WB083-N-3 south” revealed that portions of the epoxy near the top of the anchor and on a corresponding area within the anchor hole had a darker brown appearance, as shown in figure 23. Pressing the darkest of this material, shown in figure 24, with the point of a dental pick showed that the darker material was of a hardness that was similar to the hardness of the epoxy elsewhere on the “WB083-N-3 south” anchor and other core samples. A smaller area of dark brown discolored epoxy was noted on the anchor approximately opposite from the portion of the anchor shown in figure 23. It was also noted that the color of the void surfaces on core sample “WB083-N-3 south” was slightly darker and was slightly more brown in color when compared to the remainder of the core samples.

3.5 WB083-N-1 north

The two portions of core sample “WB083-N-1 north” were not sectioned because each piece was relatively short. Also, the lower portion contained multiple pieces of rebar, as shown in figure 5. Substantial void areas could be seen in the hole for this sample, and loosely adhering epoxy was removed from the hole for ease of examination. Figure 25 shows a view looking down into the hole in the lower portion of the sample after removal of the epoxy.

4. PERCENT VOID AREA

The percentage of the cylindrical area between the anchor and the hole surface occupied by void areas was estimated for the five core samples retained by the Safety Board. This estimation was done by mapping out the void areas on graph paper and then graphically measuring the area. The void percent was in relation to the 5-inch-length of the hole and did not include the smooth surface of the epoxy created where the epoxy was in contact with the red plastic cap. On core sample “WB084-S-3 south”, the separation surface inside the #11 rebar was evaluated by direct examination without further cutting of the bar. However, the #11 rebar at the top of core sample “WB083-N-3 north” was cut to

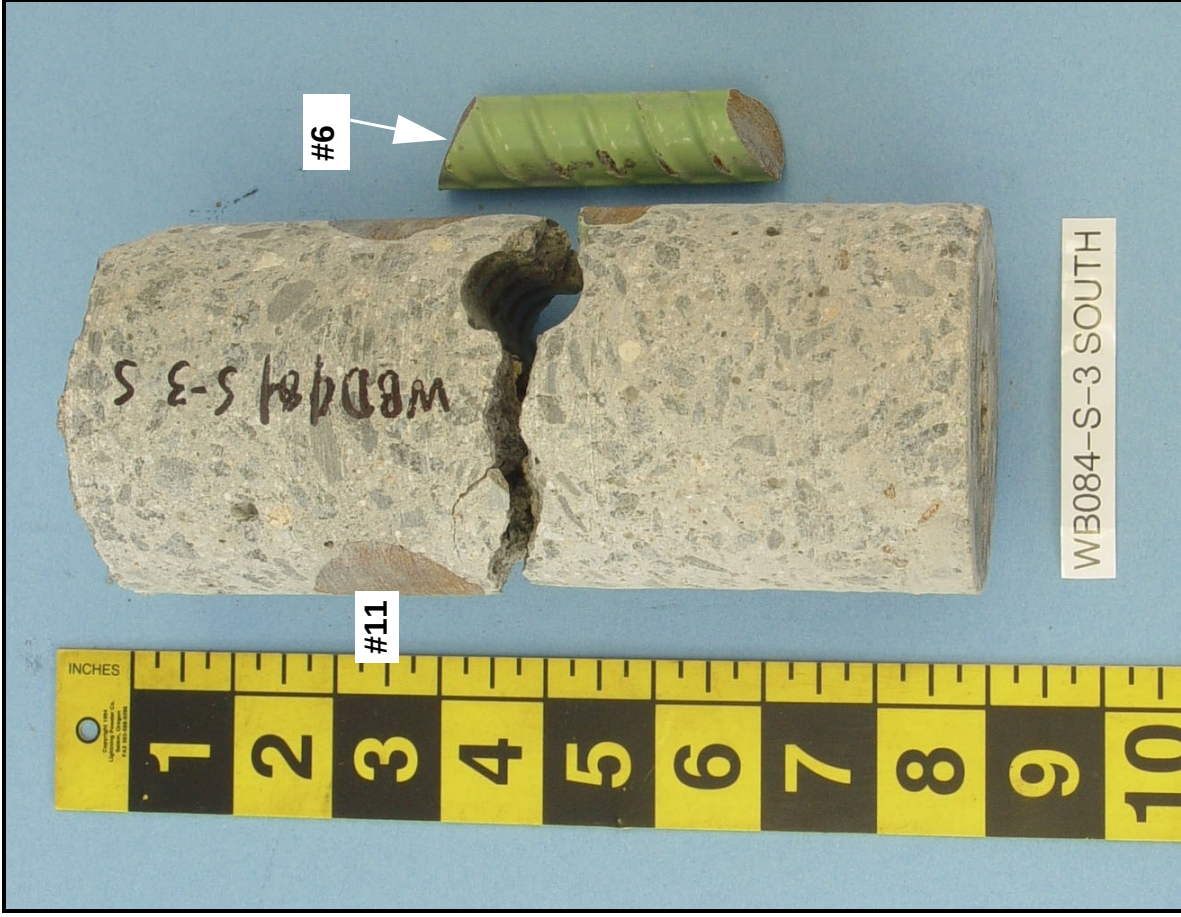
allow more detailed examination of the portion of the anchor hole passing through the bar. The following table shows the measured void area percent as well as a percent of the 5-inch cylindrical hole length occupied by the cap, and the percent of the bond area evaluated. The final column in the table indicates if the anchor had been displaced downward.

Core Sample Void Area				
Anchor Position	Void Area	Cap Area	Pct. Evaluated	Anchor Displaced?
WB084-S-3 south	21 %	13 %	100 %	No
WB083-S-8 north	10 %	12 %	100 %	No
WB083-N-3 north	25 %	12.5 %	100 %	No
WB083-N-3 south	15 %	12.5 %	90 %	Yes
WB083-N-1 north	27 %	11 %	100 %	No

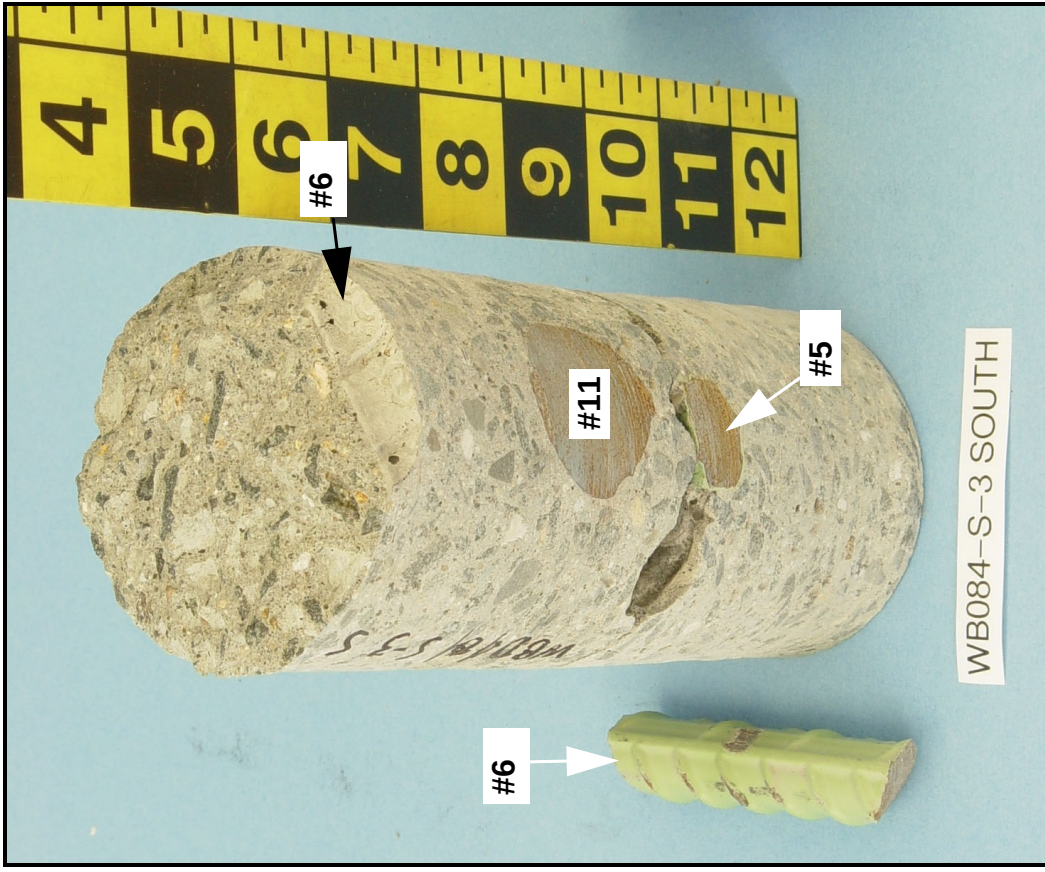
No attempt was made to measure the volume percent of the void areas. Also, it should be noted that the actual percentage of void areas could be higher than the percentages listed in the table, since it is possible that evidence of voids could have been lost or not exposed for the following reasons:

- Movement of the anchor, either while in service ("WB083-N-3 south") or during removal after the accident ("WB084-S-3 south" and "WB083-N-1 north"), could have obscured void features.
- The fracture of the epoxy during opening of the core samples might not have progressed through all of the void areas ("WB083-S-8 north" and "WB083-N-3 north").

James F. Wildey II
Supervisory Metallurgist

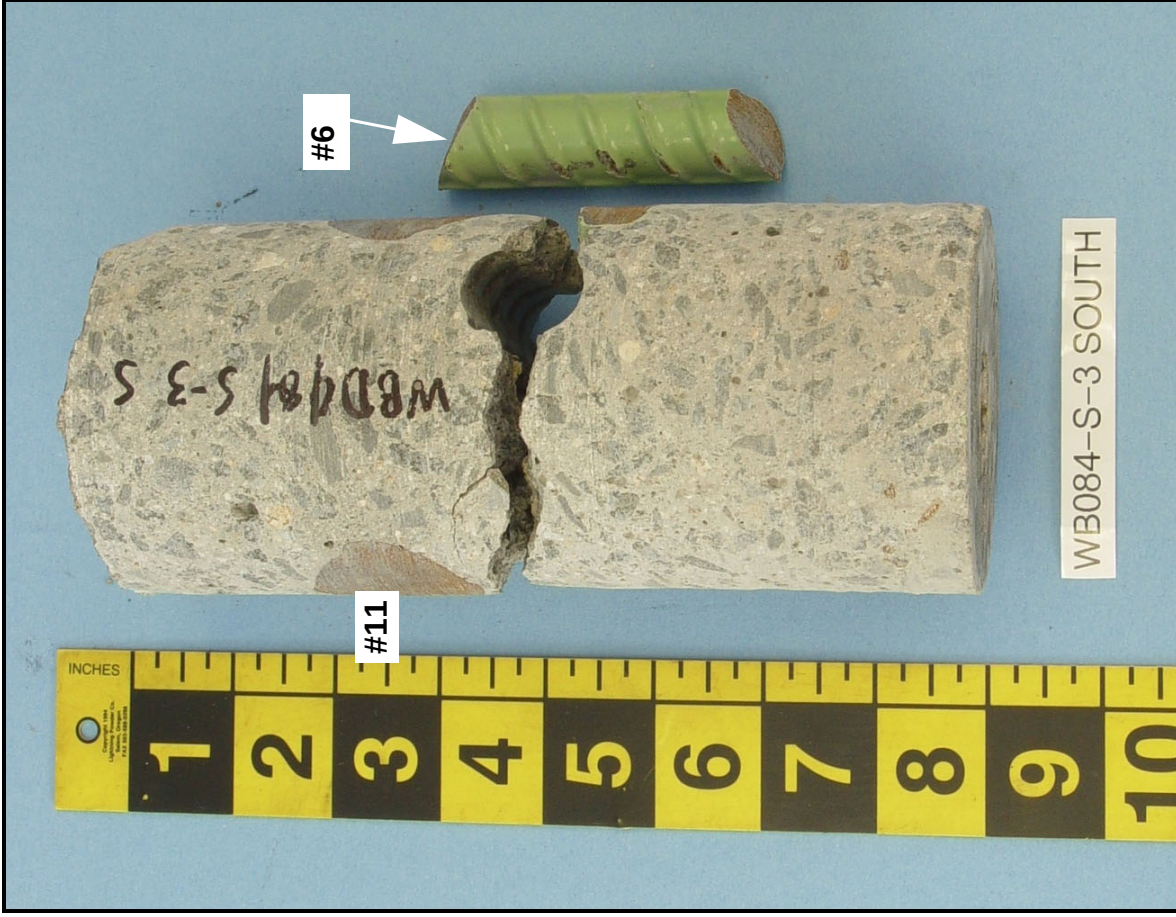


ImageNo:0608A00414, Project No:2006080010

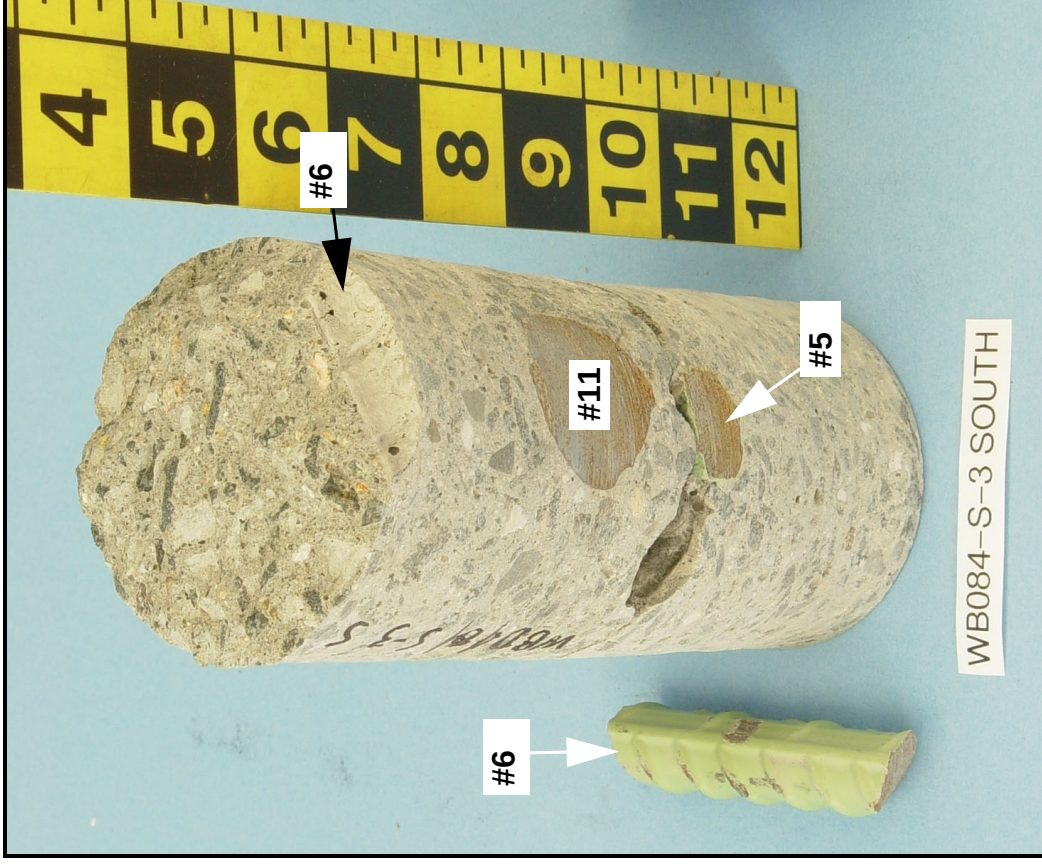


ImageNo: 0608A00419, Project No:2006080010

Figure 1. Overall side view (left photo) and oblique view (right photo) of core sample "WB084-S-3 south". The locations and sizes of rebar are indicated.

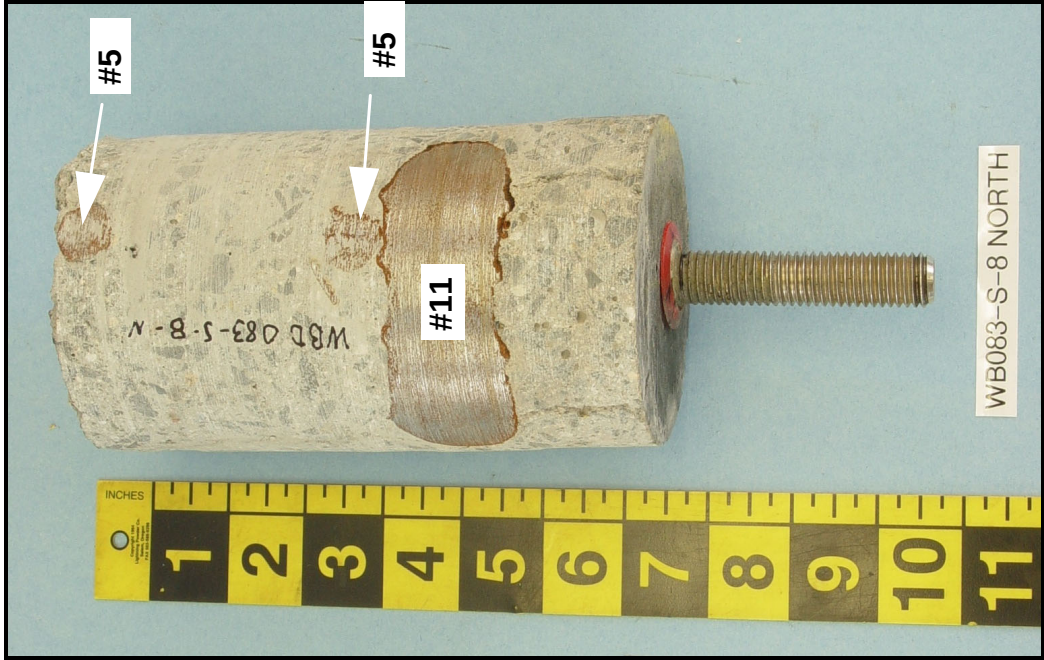


ImageNo:0608A00414, Project No:2006080010

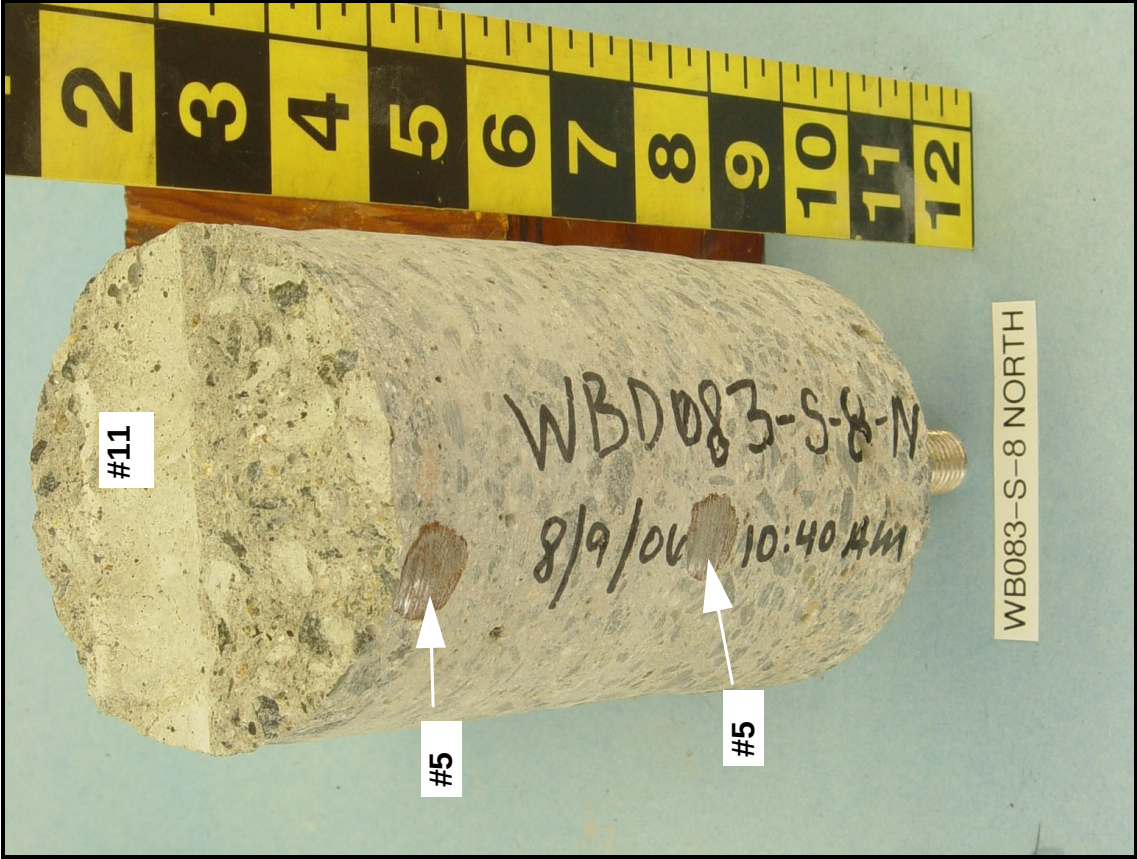


ImageNo: 0608A00419, Project No:2006080010

Figure 1. Overall side view (left photo) and oblique view (right photo) of core sample "WB084-S-3 south". The locations and sizes of rebar are indicated.

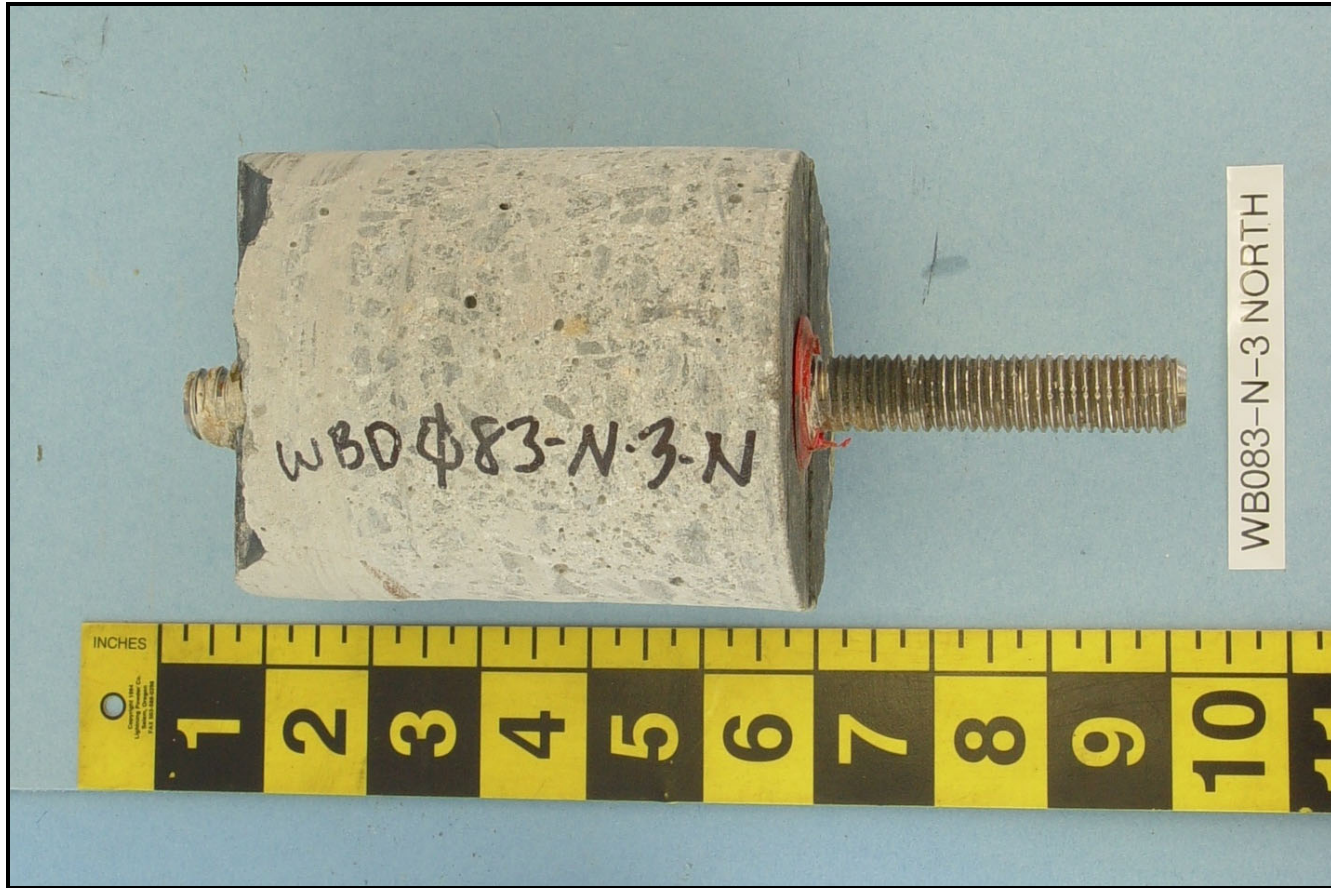


ImageNo:0608A00418, Project No:2006080010

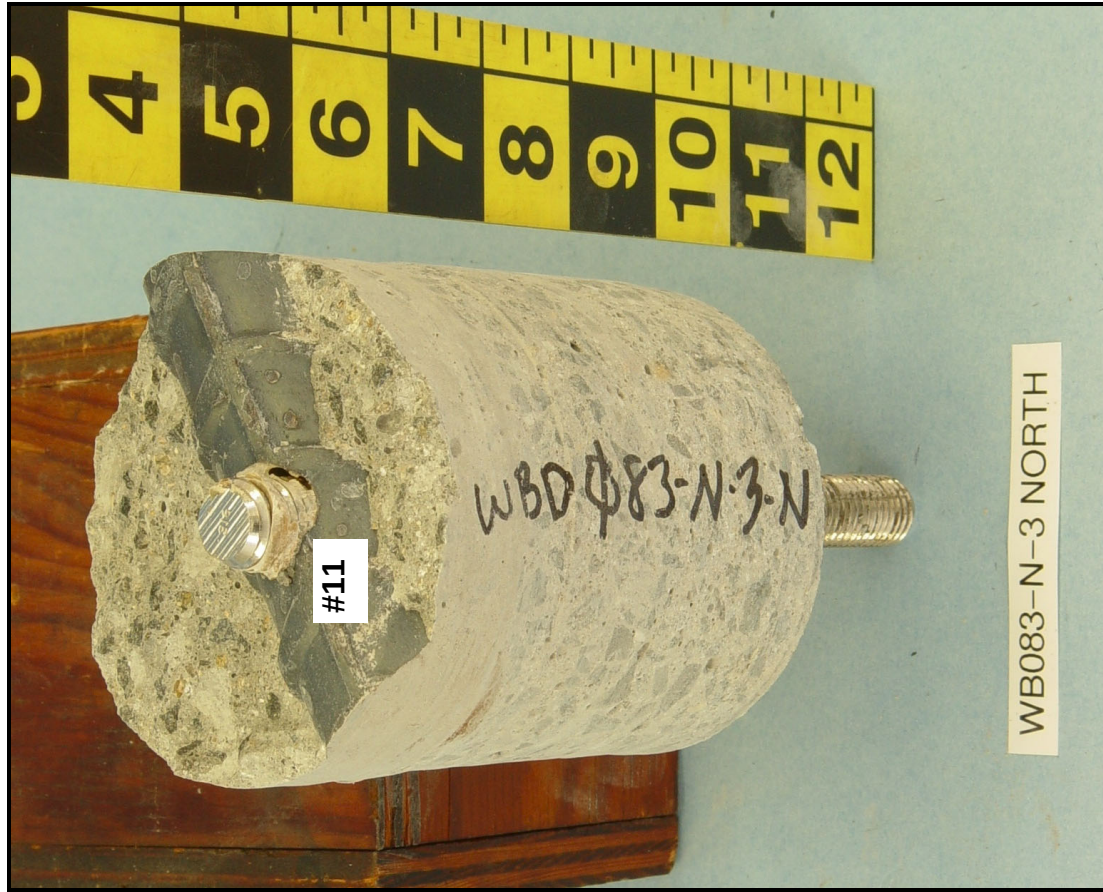


ImageNo: 0608A00426, Project No:2006080010

Figure 2. Overall side view (left photo) and oblique view (right photo) of the core sample "WB083-S-8 north". The locations and sizes of rebar are indicated.



ImageNo:0608A00416, Project No:2006080010

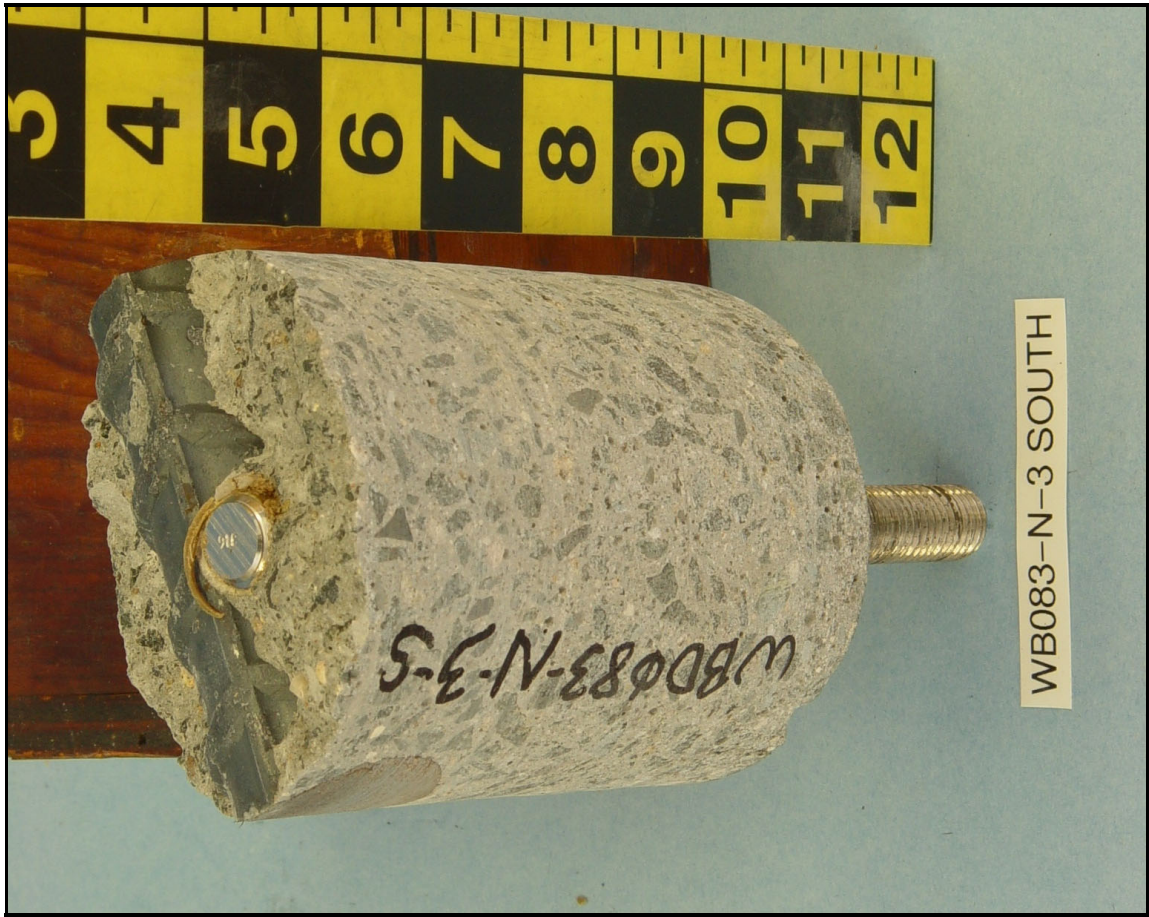


ImageNo: 0608A00424, Project No:2006080010

Figure 3. Overall side view (left photo) and oblique view (right photo) of the core sample "WB083-N-3 north". The location and size of rebar are indicated.

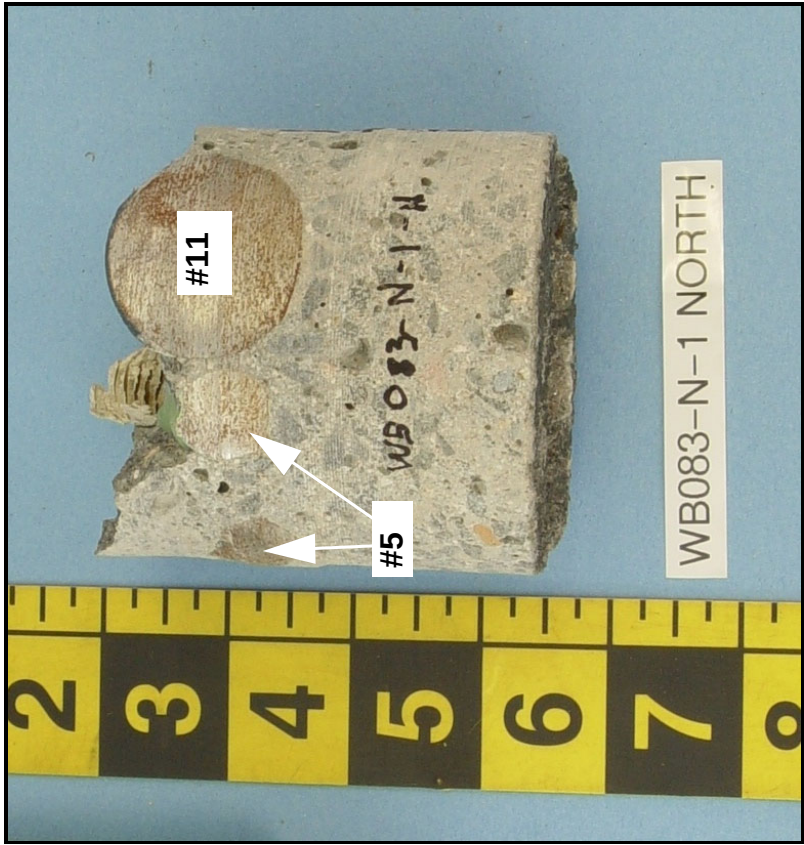


ImageNo: 0608A00417

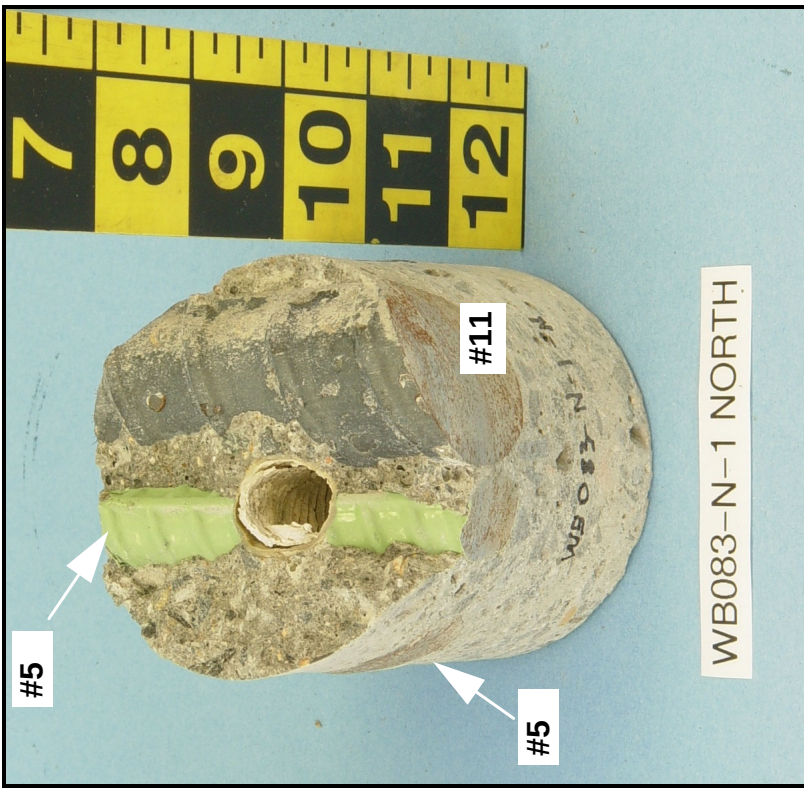


ImageNo: 0608A00425, Project No: 2006080010

Figure 4. Overall side view (left photo) and oblique view (right photo) of the core sample "WB083-N-3 south". The location and size of rebar are indicated.



ImageNo:0608A00415, Project No:2006080010



ImageNo: 0608A00420, Project No:2006080010

Figure 5. Overall side view (left photo) and oblique view (right photo) of the lower portion of core sample "WB083-N-1 north". The locations and sizes of rebar are indicated. The upper portion of this sample is not shown.



ImageNo:0608A01073, Project No:2006080010

Figure 6. X-ray image at 0 degrees of core sample "WB083-S-8 north", with the arrow indicating what appeared to be an irregular void area above the upper end of the anchor.



ImageNo: 0608A00868, Project No:2006080010

Figure 7. Lower portion of core sample "WB084-S-3 south" after longitudinal cutting. A closer view of the piece on the right is shown in figure 8.



ImageNo:0608A00891, Project No:200 10 mm

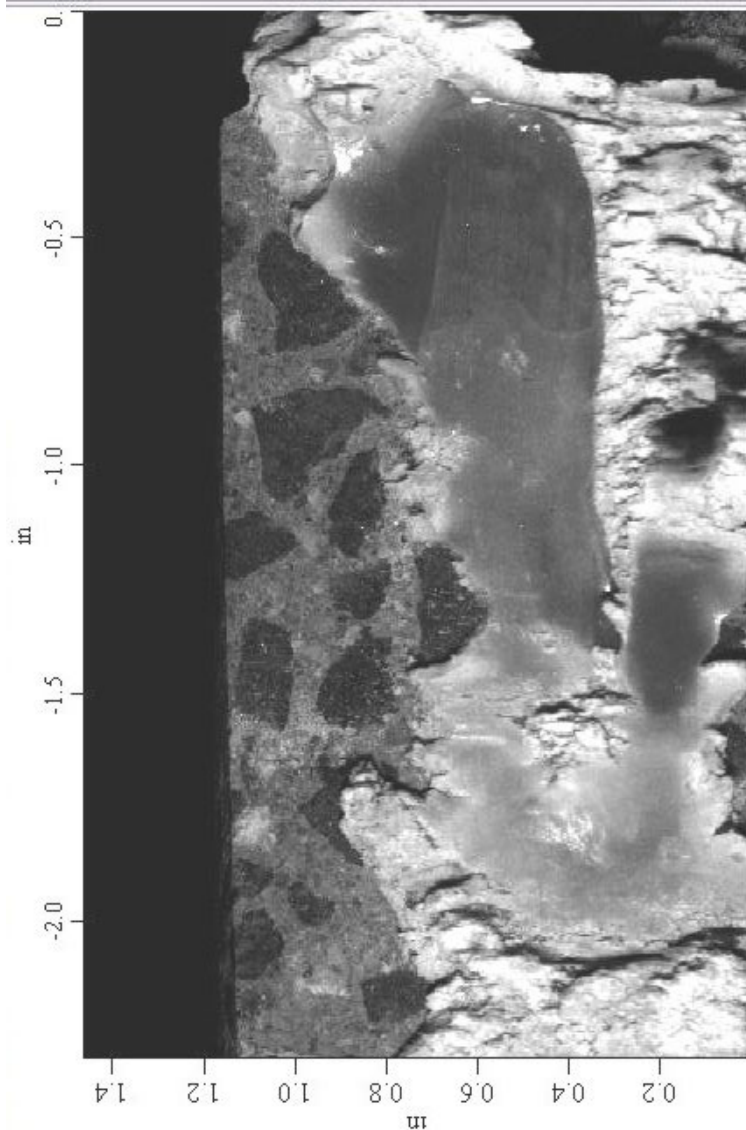


Figure 9. Laser-generated image from Laser Techniques Company showing the upper portion of the area shown in figure 8 (from core sample "WB084-S-3 south").

Figure 8. Composite photograph with a closer view of the epoxy remaining in one half of the drilled hole in the lower portion of core sample "WB084-S-3 south". Dashed lines outline void areas. The plastic cap extended up inside the hole to the position indicated.



Figure 10. Overall view of the anchor from position "WB084-S-3 south", showing the smooth but axially compressed epoxy on this side of the anchor.



ImageNo:0609A00519, Project No:2006080010

Figure 11. Upper portion of core sample "WB084-S-3 south" after cutting near the upper side of the #11 rebar in order to expose the upper end of the drilled hole. The arrows indicate the two sides of a void in the epoxy.



Figure 12. Closer view of the upper end of the drilled hole from core sample "WB084-S-3 south" (from the piece shown on the left in figure 9) after baking and removal of the epoxy.

ImageNo: 0609A00530, Project No:2006080010

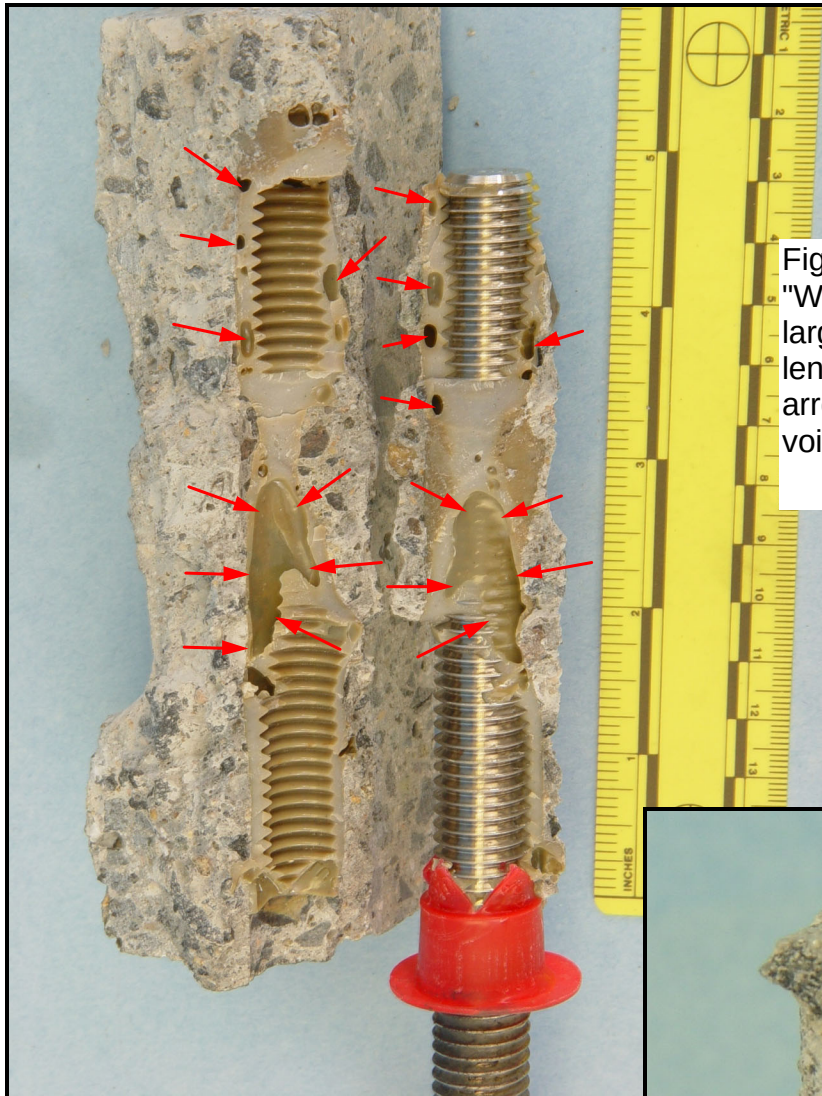


Figure 13. Opened core sample "WB083-S-8 north" showing a large void midway along the length of the anchor (outlined by arrows) and several smaller voids (individual arrows).

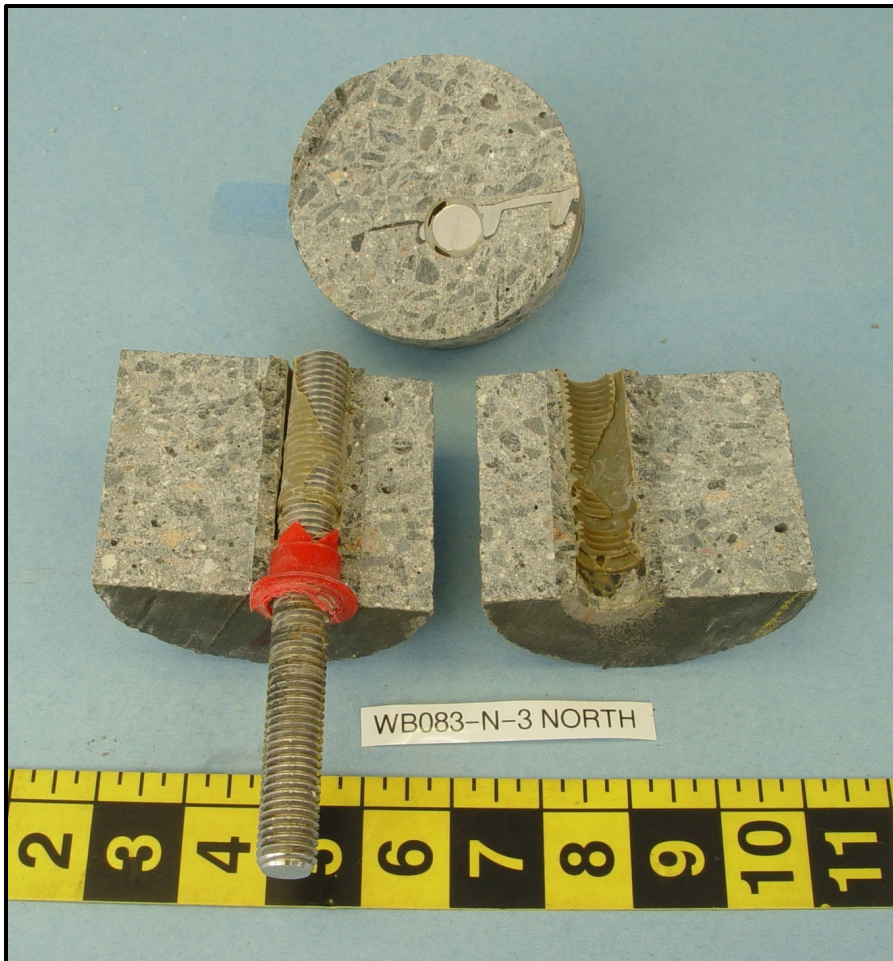
ImageNo:0609A00567, Project No:2006080010

Figure 14. Opened core sample "WB083-S-8 north" (before final removal of the anchor) showing a large void and several smaller voids at and above the upper end of the anchor.



ImageNo: 0609A00266, Project No:2006080010

10 mm



ImageNo:0609A00268, Project No:2006080010

Figure 15. Overall view of core sample "WB083-N-3 north" after transverse and longitudinal sectioning.

Figure 16. Views of the two halves of core sample "WB083-N-3 north" below the #11 rebar. Smooth areas without thread impressions are a single continuous void.



↑
UP

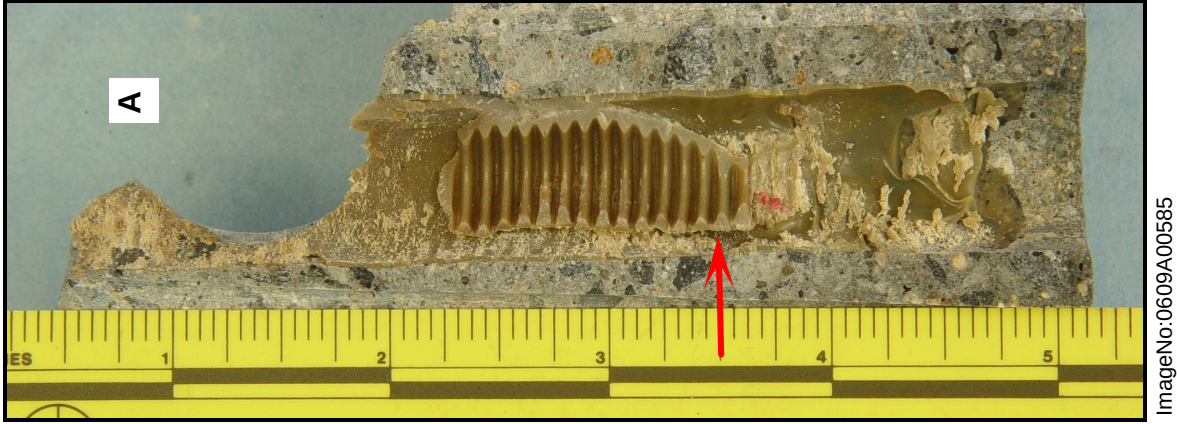
ImageNo: 0609A00580, Project No:20060800:



Figure 17. Transverse cut just below the #11 rebar on core sample "WB083-N-3 north". Arrows indicate void areas.



Figure 18. Core sample "WB083-N-3 south" after cutting.



↑
UP

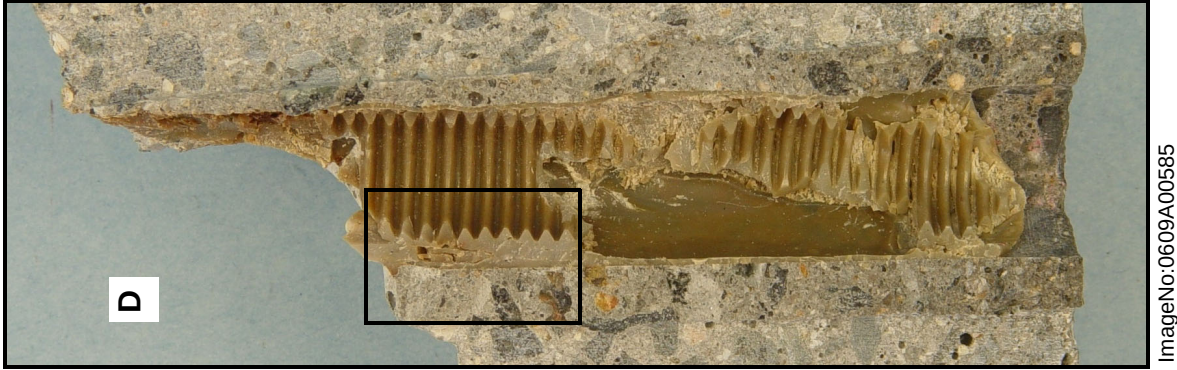
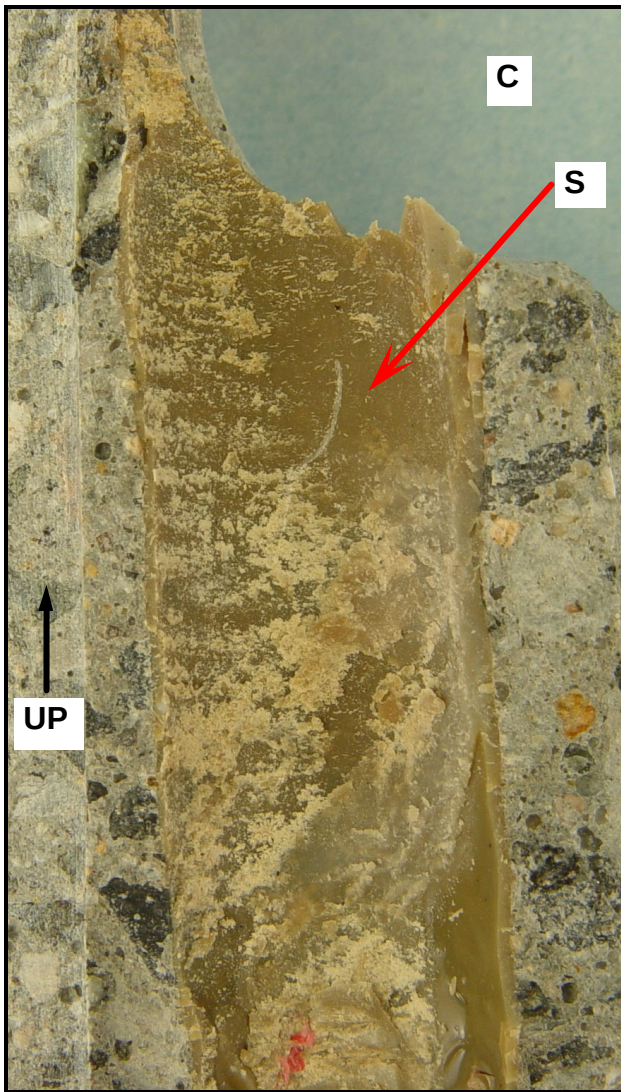


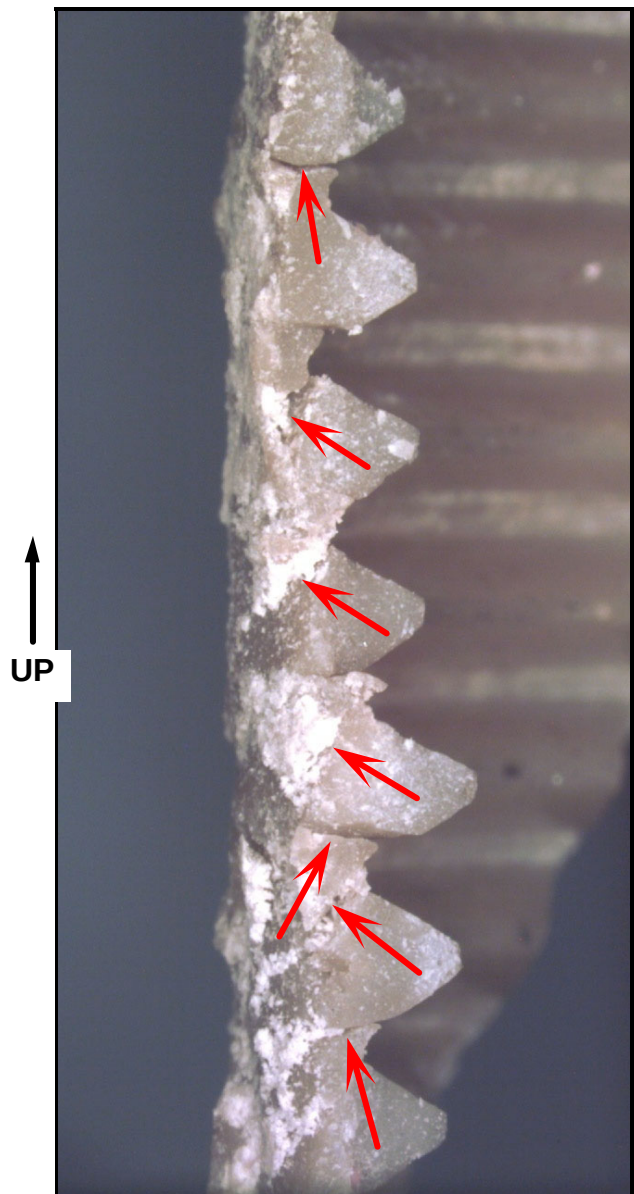
Figure 19. Views of core sample "WB083-N-3 south" after removal of the #11 rebar and splitting in half. Two left photographs were taken before epoxy was chipped away to reveal the extent of void areas, and two right photographs were taken after epoxy was chipped away.



ImageNo:0609A00694, Project No:2006080010

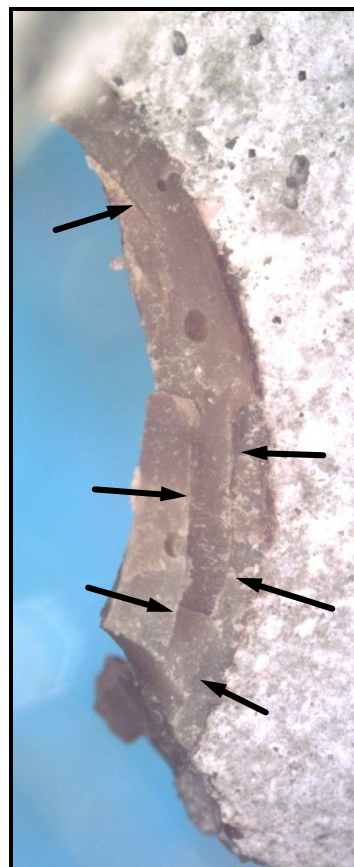
Figure 20. Closer view of the epoxy separation surface shown in figure 19, photograph "C". The center of an area with smooth features is indicated by arrow "S", and textured bands are visible on the left side of the photograph.

Figure 21. Edge view of a piece of the epoxy removed from core sample "WB083-N-3 south" after it was split in half. The arrow in photograph "A" of figure 19 indicates the approximate location on the removed piece where this photograph was taken as well as the direction of viewing of this photograph. The red arrows in this photograph indicate some of the cracks intersecting the thread crowns.

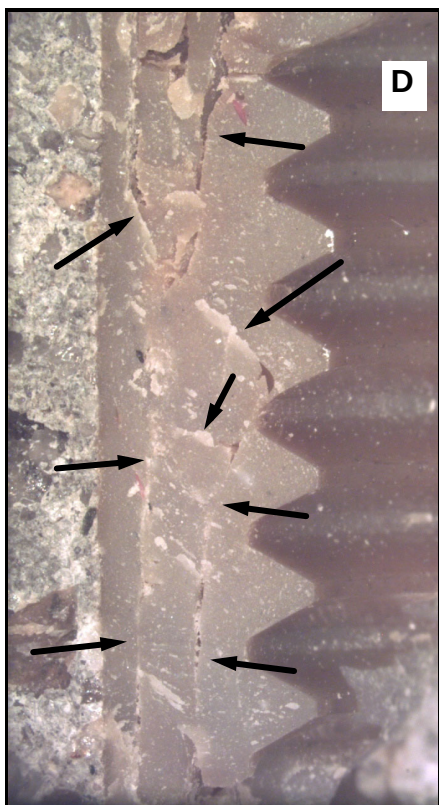


ImageNo: 0609A00695, Project No:20060 | 2 mm

Figure 22. Views of the fractured epoxy remaining in the split halves of core sample "WB083-N-3 south". Lower left image is a closer view of the area in the box in figure 19, image "D". Lower right image is a closer view of the area in the box in figure 19, image "C". Upper right image is a view looking down on the area shown in the lower right. Arrows in all three views indicate cracks in the epoxy.



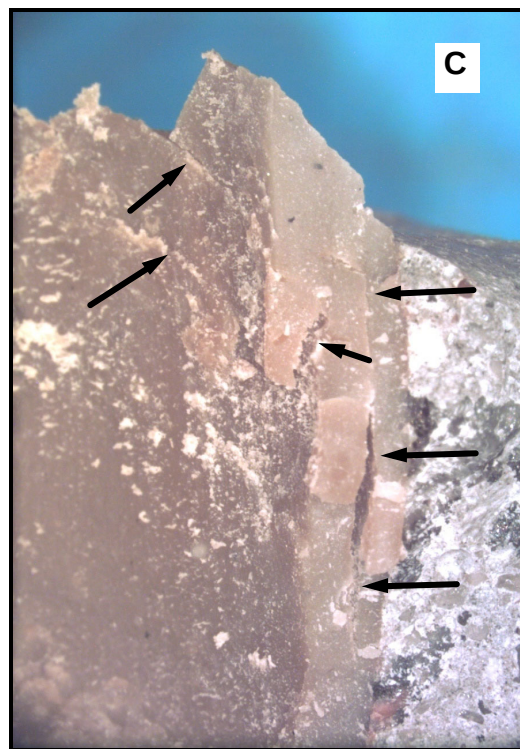
ImageNo:0609A00652 1 mm



ImageNo:0609A00652

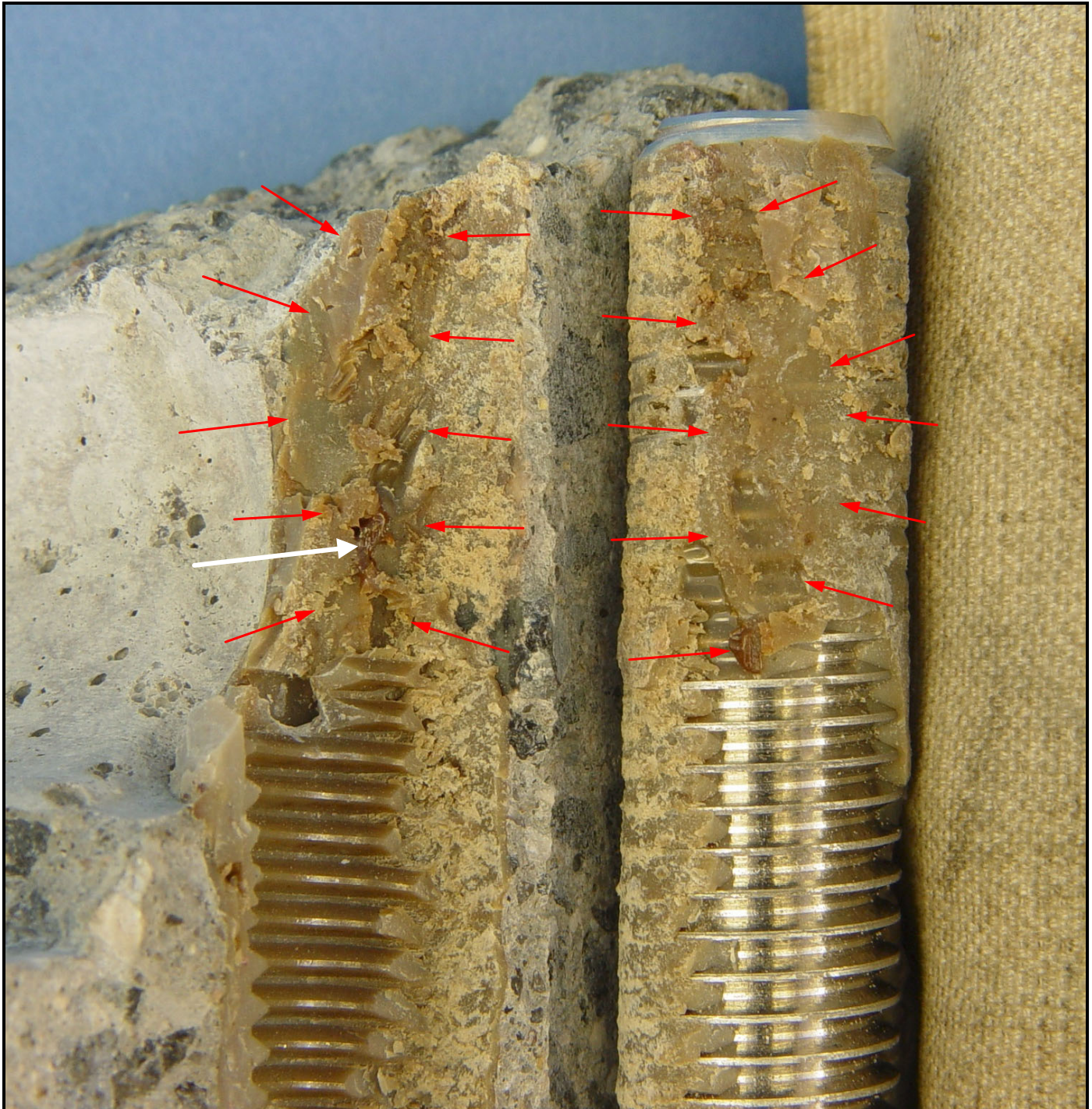
2 mm

↑
UP



ImageNo: 0609A00654

2 mm



ImageNo:0610A00050, Project No:2006080010

10 mm

Figure 23. View of the upper end of the anchor (on the right) and a portion of the upper end of the anchor hole (on the left) from position "WB083-N-3 south", showing the darker brown appearance of the epoxy in the area outlined by the red arrows.



ImageNo:0610A00090, Project No:2006080010

5 mm

Figure 24. Closer view of the darkest discolored area, located on the epoxy attached to the anchor hole on core sample "WB083-N-3 south", at the position indicated by the larger white arrow in figure 23.



ImageNo: 0609A00705, Project No:2006080010

10 mm

Figure 25. View looking down inside the anchor hole on the lower portion of core sample "WB083-N-1 north" showing a major void area. Photograph was taken after removal of loose epoxy. Compare to figure 5.